

9	C	When the string just slackens, the tension in the string is zero. The component of the weight acting in the radial direction provides for the centripetal force. $mg\cos\theta = \frac{mv_B^2}{r}$ $g\cos\theta = \frac{v_B^2}{r}$ $\theta = \cos^{-1}\left(\frac{v_B^2}{rg}\right) = \cos^{-1}\left[\frac{2.2^2}{(1.5)(9.81)}\right]$ $= 71^\circ$
10	D	